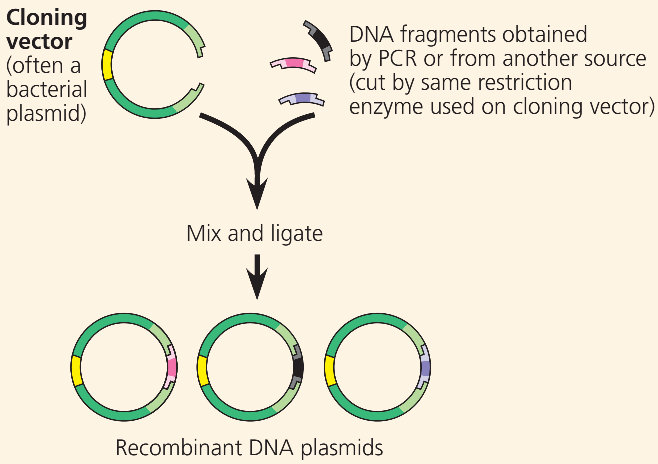
- To clone a eukaryotic gene:



Recombinant **plasmids** are returned to host cells, each of which divides to form a clone of cells.
- Expressing cloned eukaryotic genes in bacterial host cells poses several technical difficulties. The use of cultured eukaryotic cells as host cells, coupled with appropriate **expression vectors**, helps avoid these problems.

? *Describe how the process of gene cloning results in a cell clone containing a recombinant plasmid.*

CONCEPT **20.2**

Biologists use DNA technology to study gene expression and function *(pp. 423–428)*

- Several techniques use hybridization of a **nucleic acid probe** to detect the presence of specific mRNAs.
- ***In situ* hybridization** and **RT-PCR** can detect the presence of a given mRNA in a tissue or an RNA sample, respectively.
- Sets of genes co-expressed by a group of cells can be detected by **RNA sequencing (RNA-seq)**—sequencing the **cDNAs** corresponding to mRNAs from the cells. **DNA microarrays** are also used for this purpose.
- For a gene of unknown function, experimental inactivation of the gene (a gene knockout) and observation of the resulting phenotypic effects can provide clues to its function. The CRISPR-Cas9 system allows researchers to edit genes in living cells in a specific, desired way. The new alleles can be altered so that they are inherited in a biased way through a population (**gene drive**).
- In humans, **genome-wide association studies** identify and use **single nucleotide polymorphisms (SNPs)** as genetic markers for alleles that are associated with particular conditions.

? *What useful information is obtained by detecting expression of specific genes?*

CONCEPT **20.3**

Cloned organisms and stem cells are useful for basic research and other applications *(pp. 428–432)*

- The question of whether all the cells in an organism have the same genome prompted the first attempts at organismal cloning.
- Single differentiated cells from plants are often **totipotent**: capable of generating all the tissues of a complete new plant.
- Transplantation of the nucleus from a differentiated animal cell into an enucleated egg can sometimes give rise to a new animal.
- Certain embryonic **stem cells** (ES cells) from animal embryos and particular adult stem cells from adult tissues can reproduce and differentiate both in the lab and in the organism, offering the potential for medical use. ES cells are **pluripotent** but difficult to acquire. Induced pluripotent stem (iPS) cells resemble ES cells in their capacity to differentiate; they can be generated by reprogramming differentiated cells. iPS cells hold promise for medical research and regenerative medicine.

? *Describe how, using mice, a researcher could carry out (1) organismal cloning, (2) production of ES cells, and (3) generation of iPS cells, focusing on how the cells are reprogrammed. (The procedures are basically the same in humans and mice.)*

CONCEPT **20.4**

The practical applications of DNA-based biotechnology affect our lives in many ways *(pp. 433–439)*

- DNA technology, including the analysis of genetic markers such as SNPs, is increasingly being used in the diagnosis of genetic and other diseases and in personal genome analysis. Personalized medicine offers potential for an individual minimizing their known risk for a disease, as well as better treatment of genetic disorders or cancers. **Gene therapy** or gene editing with the CRISPR-Cas9 system may also lead to permanent cures. DNA technology is used with cell cultures in the large-scale production of protein hormones and other proteins with therapeutic uses. Some therapeutic proteins are being produced in **transgenic** "pharm" animals.
- Analysis of genetic markers such as **short tandem repeats (STRs)** in DNA isolated from tissue or body fluids found at crime scenes leads to a **genetic profile**. Use of genetic profiles can provide definitive evidence that a suspect is innocent or strong evidence of guilt. Such analysis is also useful in parenthood disputes and in identifying the remains of crime victims.
- Genetically engineered microorganisms can be used to extract minerals from the environment or degrade various types of toxic waste materials.
- The aims of developing transgenic plants and animals are to improve agricultural productivity and food quality.
- The potential benefits of genetic engineering must be carefully weighed against the potential for harm to humans or the environment.

? *What factors affect whether a given genetic disease would be a good candidate for successful gene therapy?*

TEST YOUR UNDERSTANDING

For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

1. In DNA technology, the term *vector* can refer to
(A) the enzyme that cuts DNA into restriction fragments.
(B) the sticky end of a DNA fragment.
(C) a SNP marker.
(D) a plasmid used to transfer DNA into a living cell.

2. Which of the following tools of DNA technology is incorrectly paired with its use?
(A) electrophoresis—separation of DNA fragments
(B) DNA ligase—cutting DNA, creating sticky ends of restriction fragments
(C) DNA polymerase—polymerase chain reaction to amplify sections of DNA
(D) reverse transcriptase—production of cDNA from mRNA